STAT 432 HW7: LDA vs. QDA

STAT 432

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0.1 Problem setup

We compare ordinary linear discriminant analysis (LDA) and quadratic discriminant analysis (QDA) for two equally likely classes and 10 predictors. Given class 1, the first two predictors have mean $(-1, 0)^T$ and covariance $\begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$. Given class 2, their mean is $(1, 0)^T$ and covariance $\begin{pmatrix} 1 & -\rho \\ -\rho & 1 \end{pmatrix}$. The other eight predictors are independent standard normal noise, independent of the first two and of the class label.

We use $\rho \in \{0, 0.2, 0.4, 0.6, 0.8, 0.9\}$, 100 training observations per class, and 1,000 test observations per class. The same generated training and test observations are used for LDA and QDA within every comparison. The random seed is **4321**.

```
set.seed(4321)

rhos <- c(0, 0.2, 0.4, 0.6, 0.8, 0.9)
n_train <- 100
n_test <- 1000
class_prior <- c(0.5, 0.5)

draw_class <- function(n, mean_xy, covariance_xy) {
  xy <- MASS::mvrnorm(n, mu = mean_xy, Sigma = covariance_xy)
  noise <- matrix(rnorm(n * 8), nrow = n, ncol = 8)
  cbind(xy, noise)
}

generate_data <- function(rho, n_per_class) {
  covariance_1 <- matrix(c(1, rho, rho, 1), nrow = 2)
  covariance_2 <- matrix(c(1, -rho, -rho, 1), nrow = 2)

  x <- rbind(
    draw_class(n_per_class, c(-1, 0), covariance_1),
    draw_class(n_per_class, c(1, 0), covariance_2)
  )
  y <- factor(rep(c("1", "2"), each = n_per_class), levels = c("1", "2"))
  list(x = x, y = y)
}

compare_methods <- function(train, test) {
  lda_fit <- MASS::lda(
    x = train$x, grouping = train$y, prior = class_prior
  )
  qda_fit <- MASS::qda(
```

```
    x = train$x, grouping = train$y, prior = class_prior
  )

  lda_prediction <- predict(lda_fit, newdata = test$x)$class
  qda_prediction <- predict(qda_fit, newdata = test$x)$class

  c(
    LDA = mean(lda_prediction != test$y),
    QDA = mean(qda_prediction != test$y)
  )
}
```

0.2 1. One dataset for each value of rho

For each ρ, generate one training set and one independent test set, fit both methods on that training set, and calculate test misclassification rates.

```
single_results <- lapply(rhos, function(rho) {
  train <- generate_data(rho, n_train)
  test <- generate_data(rho, n_test)
  errors <- compare_methods(train, test)
  data.frame(rho = rho, LDA = errors["LDA"], QDA = errors["QDA"])
}) |>
  bind_rows()

knitr::kable(
  single_results,
  digits = 4,
  col.names = c("rho", "LDA test error", "QDA test error"),
  caption = "Test misclassification rates for one dataset at each rho."
)
```

Table 1: Test misclassification rates for one dataset at each rho.

	rho	LDA test error	QDA test error
LDA...1	0.0	0.1655	0.1900
LDA...2	0.2	0.1660	0.1935
LDA...3	0.4	0.1805	0.1905
LDA...4	0.6	0.1620	0.1625
LDA...5	0.8	0.1810	0.1480
LDA...6	0.9	0.1835	0.1060

In this one simulated run, LDA has lower error at the smallest values of ρ, while QDA has lower error at $\rho = 0.8$ and $\rho = 0.9$. A single simulation is variable, so the repeated experiment in Question 2 gives a more stable comparison.

0.3 2. Average error over 50 repetitions

At each fixed ρ, we generate 50 new training and test sets. Each repetition uses identical data for the two methods so their errors are comparable.

```r
n_repetitions <- 50

repeated_results <- lapply(rhos, function(rho) {
  replicate_errors <- replicate(n_repetitions, {
    train <- generate_data(rho, n_train)
    test <- generate_data(rho, n_test)
    compare_methods(train, test)
  })

  data.frame(
    rho = rho,
    repetition = seq_len(n_repetitions),
    LDA = replicate_errors["LDA", ],
    QDA = replicate_errors["QDA", ]
  )
}) |>
  bind_rows()

average_errors <- repeated_results |>
  group_by(rho) |>
  summarise(
    LDA = mean(LDA),
    QDA = mean(QDA),
    .groups = "drop"
  )

average_errors_long <- average_errors |>
  pivot_longer(
    cols = c(LDA, QDA),
    names_to = "Method",
    values_to = "Average test error"
  )

ggplot(
  average_errors_long,
  aes(x = rho, y = `Average test error`, color = Method, group = Method)
) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  scale_x_continuous(breaks = rhos) +
  labs(
    x = expression(rho),
    y = "Average test misclassification rate",
    title = "LDA and QDA test error over 50 repetitions",
    color = "Method"
  ) +
  theme_minimal(base_size = 12)
```

```
knitr::kable(
  average_errors,
  digits = 4,
  col.names = c("rho", "LDA average error", "QDA average error"),
  caption = "Average test misclassification rate over 50 repetitions."
)
```

Table 2: Average test misclassification rate over 50 repetitions.

rho	LDA average error	QDA average error
0.0	0.1700	0.1953
0.2	0.1699	0.1953
0.4	0.1693	0.1908
0.6	0.1747	0.1780
0.8	0.1691	0.1342
0.9	0.1663	0.0983

The averages show LDA performing better for small ρ, with QDA's average error decreasing substantially for large ρ. The methods become similar around $\rho = 0.6$; QDA is better at $\rho = 0.8$ and $\rho = 0.9$.

0.4 3. Interpretation

When $\rho = 0$, both classes have the same covariance matrix for all 10 predictors. LDA's common-covariance assumption is therefore correct. For small ρ, the class covariances remain relatively similar, so LDA's simpler model generally has lower estimation variance. QDA fits a separate covariance matrix in each class; at small ρ, that added flexibility offers little bias reduction and can increase test error.

As ρ grows, the first two predictors have positive correlation in class 1 and negative correlation in class 2. The true class covariance matrices therefore differ more strongly. LDA pools those covariance structures and cannot represent the class-specific correlations, while QDA can. At high ρ, QDA's reduction in model bias outweighs its higher variance, and its average test error becomes lower. This is the bias-variance trade-off.

The eight noise predictors have no population-level relationship with the class label, but estimating their means and covariances from a finite training sample still adds sampling variability. In particular, QDA estimates a full 10-by-10 covariance matrix separately for each class, while LDA estimates one pooled covariance matrix. Noisy covariance estimates can introduce spurious terms into the fitted decision boundary, so the extra predictors can worsen generalization—especially for QDA with only 100 observations per class.

Conclusion: LDA is a strong choice when the class covariance matrices are similar; QDA can perform better when those matrices differ substantially. The noise predictors add no useful signal, but their estimation cost can increase test error.